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Sheet with progress tracking: https://thita.ai/dsa-patterns-sheet

Beginner's order: https://thita.ai/dsa-patterns-sheet?order=beginner

Theory for dsa : https://thita.ai/dashboard/learning-path/dsa

Discord community: https://discord.gg/zxywjSuvDT

Tip: you dont need to solve all questions, try to solve one medium and 1 hard per each pattern and see if others you could understand through them.

Pattern	**Problems**
I. Two Pointer Patterns	
Pattern 1: Two Pointers - Converging (Sorted Array Target Sum)	11. Container With Most Water, 15. 3Sum, 16. 3Sum Closest, 18. 4Sum, 167. Two Sum II - Input Array Is Sorted, 349. Intersection of Two Arrays, 881. Boats to Save People, 977. Squares of a Sorted Array, 259. 3Sum Smaller
Pattern 2: Two Pointers - Fast & Slow (Cycle Detection)	141. Linked List Cycle, 202. Happy Number, 287. Find the Duplicate Number, 392. Is Subsequence,
Pattern 3: Two Pointers - Fixed Separation (Nth Node from End)	19. Remove Nth Node From End of List, 876. Middle of the Linked List, 2095. Delete the Middle Node of a Linked List
Pattern 4: Two Pointers - In-place Array Modification	26. Remove Duplicates from Sorted Array, 27. Remove Element, 75. Sort Colors, 80. Remove Duplicates from Sorted Array II, 283. Move Zeroes, 443. String Compression, 905. Sort Array By Parity, 2337. Move Pieces to Obtain a String, 2938. Separate Black and White Balls
Pattern 5: Two Pointers - String Comparison with Backspaces	844. Backspace String Compare, 1598. Crawler Log Folder , 2390. Removing Stars From a String
Pattern 6: Two Pointers - Expanding From Center (Palindromes)	5. Longest Palindromic Substring, 647. Palindromic Substrings
Pattern 7: Two Pointers - String Reversal	151. Reverse Words in a String, 344. Reverse String, 345. Reverse Vowels of a String, 541. Reverse String II
II. Sliding Window Patterns	
Pattern 8: Sliding Window - Fixed Size (Subarray Calculation)	346. Moving Average from Data Stream, 643. Maximum Average Subarray I, 2985. Calculate Compressed Mean, 3254. Find the Power of K-Size Subarrays I, 3318. Find X-Sum of All K-Long Subarrays I
Pattern 9: Sliding Window - Variable Size (Condition-Based)	3. Longest Substring Without Repeating Characters, 76. Minimum Window Substring, 209. Minimum Size Subarray Sum, 219. Contains Duplicate II, 424. Longest Repeating Character Replacement, 713. Subarray Product Less Than K, 904. Fruit Into Baskets, 1004. Max Consecutive Ones III, 1438. Longest Continuous Subarray With Absolute Diff Less Than or Equal to Limit, 1493. Longest Subarray of 1's After Deleting One Element, 1
Pattern 10: Sliding Window - Monotonic Queue for Max/Min	239. Sliding Window Maximum, 862. Shortest Subarray with Sum at Least K, 1696. Jump Game VI
Pattern 11: Sliding Window - Character Frequency Matching	1.TwoSum, 438. Find All Anagrams in a String, 567. Permutation in String
III. Tree Traversal Patterns (DFS & BFS)	
Pattern 12: Tree BFS - Level Order Traversal	102. Binary Tree Level Order Traversal, 103. Binary Tree Zigzag Level Order Traversal, 199. Binary Tree Right Side View, 515. Find Largest Value in Each Tree Row, 1161. Maximum Level Sum of a Binary Tree
Pattern 13: Tree DFS - Recursive Preorder Traversal	100. Same Tree, 101. Symmetric Tree, 105. Construct Binary Tree from Preorder and Inorder Traversal, 114. Flatten Binary Tree to Linked List, 226. Invert Binary Tree, 257. Binary Tree Paths, 988. Smallest String Starting From Leaf
Pattern 14: Tree DFS - Recursive Inorder Traversal	94. Binary Tree Inorder Traversal, 98. Validate Binary Search Tree, 173. Binary Search Tree Iterator, 230. Kth Smallest Element in a BST, 501. Find Mode in Binary Search Tree, 530. Minimum Absolute Difference in BST
Pattern 15: Tree DFS - Recursive Postorder Traversal	104. Maximum Depth of Binary Tree, 110. Balanced Binary Tree, 124. Binary Tree Maximum Path Sum, 145. Binary Tree Postorder Traversal, 337. House Robber III, 366. Find Leaves of Binary Tree, 543. Diameter of Binary Tree, 863. All Nodes Distance K in Binary Tree, 1110. Delete Nodes And Return Forest, 2458. Height of Binary Tree After Subtree Removal Queries
Pattern 16: Tree - Lowest Common Ancestor (LCA) Finding	235. Lowest Common Ancestor of a Binary Search Tree, 236. Lowest Common Ancestor of a Binary Tree
Pattern 17: Tree - Serialization and Deserialization	297. Serialize and Deserialize Binary Tree, 572. Subtree of Another Tree, 652. Find Duplicate Subtrees
IV. Graph Traversal Patterns (DFS & BFS)	
Pattern 18: Graph DFS - Connected Components / Island Counting	130. Surrounded Regions, 200. Number of Islands, 417. Pacific Atlantic Water Flow, 547. Number of Provinces, 695. Max Area of Island, 733. Flood Fill, 841. Keys and Rooms, 1020. Number of Enclaves, 1254. Number of Closed Islands, 1905. Count Sub Islands, 2101. Detonate the Maximum Bombs
Pattern 19: Graph BFS - Connected Components / Island Counting	542. 01 Matrix, 994. Rotting Oranges, 1091. Shortest Path in Binary Matrix
Pattern 20: Graph DFS - Cycle Detection (Directed Graph)	207. Course Schedule, 210. Course Schedule II, 802. Find Eventual Safe States, 1059. All Paths from Source Lead to Destination
Pattern 21: Graph BFS - Topological Sort (Kahn's Algorithm)	210. Course Schedule II, 269. Alien Dictionary, 310. Minimum Height Trees, 444. Sequence Reconstruction, 1136. Parallel Courses, 1857. Largest Color Value in a Directed Graph, 2050. Parallel Courses III, 2115. Find All Possible Recipes from Given Supplies, 2392. Build a Matrix With Conditions
Pattern 22: Graph - Deep Copy / Cloning	133. Clone Graph,1334. Find the City With the Smallest Number of Neighbors at a Threshold Distance, 138. Copy List with Random Pointer,1490. Clone N-ary Tree
Pattern 23: Graph - Shortest Path (Dijkstra's Algorithm)	743. Network Delay Time, 778. Swim in Rising Water, 1514. Path with Maximum Probability, 1631. Path With Minimum Effort, 1976. Number of Ways to Arrive at Destination, 2045. Second Minimum Time to Reach Destination, 2203. Minimum Weighted Subgraph With the Required Paths, 2290. Minimum Obstacle Removal to Reach Corner, 2577. Minimum Time to Visit a Cell In a Grid, 2812. Find the Safest Path in a Grid
Pattern 24: Graph - Shortest Path (Bellman-Ford / BFS+K)	787. Cheapest Flights Within K Stops,1129. Shortest Path with Alternating Colors
Pattern 25: Graph - Union-Find (Disjoint Set Union - DSU)	200. Number of Islands, 261. Graph Valid Tree, 305. Number of Islands II, 323. Number of Connected Components in an Undirected Graph, 3 547. Number of Provinces, 684. Redundant Connection, 4 721. Accounts Merge, 737. Sentence Similarity II, 947. Most Stones Removed with Same Row or Column, 952. Largest Component Size by Common Factor, 959. Regions Cut By Slashes, 1101. The Earliest Moment When Everyone Becc
Pattern 26: Strongly Connected Components (Kosaraju / Tarjan)	210. Course Schedule II, 547. Number of Provinces, 1192. Critical Connections in a Network, 2127. Maximum Employees to Be Invited to a Meeting
Pattern 27: Bridges & Articulation Points (Tarjan low-link)	1192. Critical Connections in a Network, 2360. Longest Cycle in a Graph
Pattern 28: Minimum Spanning Tree (Kruskal / Prim / DSU + heap)	1135. Connecting Cities With Minimum Cost, 1584. Min Cost to Connect All Points,1168. Optimize Water Distribution in a Village,1489. Find Critical and Pseudo-Critical Edges in Minimum Spanning Tree
Pattern 29: Bidirectional BFS (BFS optimization for known source & target	127. Word Ladder, 126. Word Ladder II, 815. Bus Routes
V. Dynamic Programming (DP) Patterns	
Pattern 30: DP - 1D Array (Fibonacci Style)	70. Climbing Stairs, 91. Decode Ways, 198. House Robber, 213. House Robber II, 337. House Robber III, 509. Fibonacci Number, 740. Delete and Earn, 746. Min Cost Climbing Stairs
Pattern 31: DP - 1D Array (Kadane's Algorithm for Max/Min Subarray)	53. Maximum Subarray, 918. Maximum Sum Circular Subarray, 2321. Maximum Score Of Spliced Array,1749. Maximum Absolute Sum of Any Subarray, 152. Maximum Product Subarray
Pattern 32 DP - 1D Array (Coin Change / Unbounded Knapsack Style)	322. Coin Change, 377. Combination Sum IV, 518. Coin Change II
Pattern 33: DP - 1D Array (0/1 Knapsack Subset Sum Style)	416. Partition Equal Subset Sum, 494. Target Sum
Pattern 34: DP - 1D Array (Word Break Style)	139. Word Break, 140. Word Break II
Pattern 35: DP - 2D Array (Longest Common Subsequence - LCS)	1143. Longest Common Subsequence, 1092. Shortest Common Supersequence,1312. Minimum Insertion Steps to Make a String Palindrome
Pattern 36: DP - 2D Array (Edit Distance / Levenshtein Distance)	72. Edit Distance, 583. Delete Operation for Two Strings, 712. Minimum ASCII Delete Sum for Two Strings
Pattern 37: DP - 2D Array (Unique Paths on Grid)	62. Unique Paths, 63. Unique Paths II, 64. Minimum Path Sum, 120. Triangle, 221. Maximal Square, 931. Minimum Falling Path Sum, 1277. Count Square Submatrices with All Ones
Pattern 38: DP - Interval DP	312. Burst Balloons, 546. Remove Boxes
Pattern 39: DP - Catalan Numbers	95. Unique Binary Search Trees II, 96. Unique Binary Search Trees, 241. Different Ways to Add Parentheses
Pattern 40: DP - Longest Increasing Subsequence (LIS)	300. Longest Increasing Subsequence, 354. Russian Doll Envelopes, 1671. Minimum Number of Removals to Make Mountain Array, 2407. Longest Increasing Subsequence II
Pattern 41: DP - Stock problems	121. Best Time to Buy and Sell Stock, 122. Best Time to Buy and Sell Stock II,123. Best Time to Buy and Sell Stock III, 188.Best Time to Buy and Sell Stock IV, 309. Best Time to Buy and Sell Stock with Cooldown
VI. Heap (Priority Queue) Patterns	
Pattern 42: Heap - Top K Elements (Selection/Frequency)	215. Kth Largest Element in an Array, 347. Top K Frequent Elements, 451. Sort Characters By Frequency, 506. Relative Ranks, 703. Kth Largest Element in a Stream, 973. K Closest Points to Origin, 1046. Last Stone Weight, 2558. Take Gifts From the Richest Pile
Pattern 43: Heap - Two Heaps for Median Finding	295. Find Median from Data Stream, 1825. Finding MK Average
Pattern 44: Heap - K-way Merge	23. Merge k Sorted Lists, 373. Find K Pairs with Smallest Sums, 378. Kth Smallest Element in a Sorted Matrix, 632. Smallest Range Covering Elements from K Lists
Pattern 45: Heap - Scheduling / Minimum Cost (Greedy with Priority Queue)	253. Meeting Rooms II, 767. Reorganize String, 857. Minimum Cost to Hire K Workers, 1642. Furthest Building You Can Reach, 1792. Maximum Average Pass Ratio, 1834. Single-Threaded CPU, 1942. The Number of the Smallest Unoccupied Chair, 2402. Meeting Rooms III
VII. Backtracking Patterns	
Pattern 46: Backtracking - Subsets (Include/Exclude)	17. Letter Combinations of a Phone Number, 77. Combinations, 78. Subsets, 90. Subsets II
Pattern 47: Backtracking - Permutations	31. Next Permutation, 46. Permutations, 60. Permutation Sequence
Pattern 48: Backtracking - Combination Sum	39. Combination Sum, 40. Combination Sum II
Pattern 49: Backtracking - Parentheses Generation	22. Generate Parentheses, 301. Remove Invalid Parentheses
Pattern 50: Backtracking - Word Search / Path Finding in Grid	79. Word Search, 212. Word Search II, 2018. Check if Word Can Be Placed In Crossword
Pattern 51: Backtracking - N-Queens / Constraint Satisfaction	37. Sudoku Solver, 51. N-Queens
Pattern 52: Backtracking - Palindrome Partitioning	131. Palindrome Partitioning,132. Palindrome Partitioning II, 1457. Pseudo-Palindromic Paths in a Binary Tree
VIII. Greedy Patterns	
Pattern 53: Greedy - Interval Merging/Scheduling	56. Merge Intervals, 57. Insert Interval, 759. Employee Free Time, 986. Interval List Intersections, 2406. Divide Intervals Into Minimum Number of Groups
Pattern 54: Greedy - Jump Game Reachability/Minimization	45. Jump Game II, 55. Jump Game
Pattern 55: Greedy - Buy/Sell Stock	121. Best Time to Buy and Sell Stock, 122. Best Time to Buy and Sell Stock II
Pattern 56: Greedy - Gas Station Circuit	134. Gas Station,2202. Maximize the Topmost Element After K Moves
Pattern 57: Greedy - Task Scheduling (Frequency Based)	621. Task Scheduler, 767. Reorganize String, 1054. Distant Barcodes
Pattern 58: Greedy - Sorting Based	455. Assign Cookies, 135. Candy, 406. Queue Reconstruction by Height, 1029. Two City Scheduling
IX. Binary Search Patterns	
Pattern 59: Binary Search - On Sorted Array/List	35. Search Insert Position, 69. Sqrt(x), 74. Search a 2D Matrix, 278. First Bad Version, 374. Guess Number Higher or Lower, 540. Single Element in a Sorted Array, 704. Binary Search, 1539. Kth Missing Positive Number
Pattern 60: Binary Search - Find Min/Max in Rotated Sorted Array	33. Search in Rotated Sorted Array, 81. Search in Rotated Sorted Array II, 153. Find Minimum in Rotated Sorted Array, 5 162. Find Peak Element, 852. Peak Index in a Mountain Array, 1095. Find in Mountain Array
Pattern 61: Binary Search - On Answer / Condition Function	410. Split Array Largest Sum, 774. Minimize Max Distance to Gas Station, 875. Koko Eating Bananas, 1011. Capacity To Ship Packages 6 Within D Days, 1482. Minimum Number of Days to Make m Bouquets, 1760. Minimum Limit of Balls in a Bag, 2064. Minimized Maximum of Products Distributed to Any Store, 2226. Maximum Candies Allocated to K Children
Pattern 62: Binary Search - Find First/Last Occurrence	34. Find First and Last Position of Element in Sorted Array, 658. Find K Closest Elements
Pattern 63: Binary Search - Median / Kth across Two Sorted Arrays	4. Median of Two Sorted Arrays, 719. Find K-th Smallest Pair Distance, 378. Kth Smallest Element in a Sorted Matrix
X. Stack Patterns	
Pattern 64: Stack - Valid Parentheses Matching	20. Valid Parentheses, 32. Longest Valid Parentheses, 921. Minimum Add to Make Parentheses Valid, 1249. Minimum Remove to Make Valid Parentheses, 1963. Minimum Number of Swaps to Make the String Balanced
Pattern 65: Stack - Monotonic Stack	402. Remove K Digits, 496. Next Greater Element I, 503. Next Greater Element II, 739. Daily Temperatures, 901. Online Stock Span, 907. Sum of Subarray Minimums, 962. Maximum Width Ramp, 1475. Final Prices With a Special Discount in a Shop, 1673. Find the Most Competitive Subsequence
Pattern 66: Stack - Expression Evaluation (RPN/Infix)	150. Evaluate Reverse Polish Notation, 224. Basic Calculator, 227. Basic Calculator II, 772. Basic Calculator III
Pattern 67: Stack - Simulation / Backtracking Helper	71. Simplify Path, 394. Decode String, 735. Asteroid Collision
Pattern 68: Stack - Min Stack Design	155. Min Stack, 895. Maximum Frequency Stack, 901. Online Stock Span
Pattern 69: Stack - Largest Rectangle in Histogram	84. Largest Rectangle in Histogram, 85. Maximal Rectangle
XI. Bit Manipulation Patterns	
Pattern 70: Bitwise XOR - Finding Single/Missing Number	136. Single Number, 137. Single Number II, 268. Missing Number, 389. Find the Difference
Pattern 71:Bitwise AND - Counting Set Bits (Hamming Weight)	191. Number of 1 Bits, 231. Power of Two, 477. Total Hamming Distance
Pattern 72: Bitwise DP - Counting Bits Optimization	338. Counting Bits, 1494. Parallel Courses II, 1442. Count Triplets That Can Form Two Arrays of Equal XOR
Pattern 73: Bitwise Operations - Power of Two/Four Check	231. Power of Two, 342. Power of Four
XII. Linked List Manipulation Patterns	
Pattern 74: Linked List - In-place Reversal	83. Remove Duplicates from Sorted List, 92. Reverse Linked List II, 206. Reverse Linked List, 25. Reverse Nodes in k-Group, 234. Palindrome Linked List, 82. Remove Duplicates from Sorted List II
Pattern 75: Linked List - Merging Two Sorted Lists	21. Merge Two Sorted Lists, Merge k Sorted Lists
Pattern 76: Linked List - Addition of Numbers	2. Add Two Numbers, 369. Plus One Linked List
Pattern 77: Linked List - Intersection Detection	160. Intersection of Two Linked Lists, 599. Minimum Index Sum of Two Lists
Pattern 78: Linked List - Reordering / Partitioning	24. Swap Nodes in Pairs, 61. Rotate List, 86. Partition List, 143. Reorder List, 328. Odd Even Linked List
XIII. Array/Matrix Manipulation Patterns	
Pattern 79: Array/Matrix - In-place Rotation	48. Rotate Image, 189. Rotate Array, 867. Transpose Matrix
Pattern 80: Array/Matrix - Spiral Traversal	54. Spiral Matrix, 59. Spiral Matrix II, 885. Spiral Matrix III, 2326. Spiral Matrix IV
Pattern 81: Array/Matrix - Set Matrix Zeroes (In-place Marking)	73. Set Matrix Zeroes, 289. Game of Life, 498. Diagonal Traverse
Pattern 82: Array - Product Except Self (Prefix/Suffix Products)	238. Product of Array Except Self, 845. Longest Mountain in Array, 2483. Minimum Penalty for a Shop
Pattern 83: Array - Plus One (Handling Carry)	66. Plus One, 43. Multiply Strings, 989. Add to Array-Form of Integer, 67. Add Binary
Pattern 84: Array - Merge Sorted Array (In-place from End)	88. Merge Sorted Array, 977. Squares of a Sorted Array
Pattern 85: Array - Cyclic Sort	41. First Missing Positive, 268. Missing Number, 287. Find the Duplicate Number, 442. Find All Duplicates in an Array, 448. Find All Numbers Disappeared in an Array
XIV. String Manipulation Patterns	
Pattern 86: String - Palindrome Check (Two Pointers / Reverse)	9. Palindrome Number, 125. Valid Palindrome, 680. Valid Palindrome II
Pattern 87: String - Anagram Check (Frequency Count/Sort)	49. Group Anagrams, 242. Valid Anagram
Pattern 88: String - Roman to Integer Conversion	13. Roman to Integer, 12. Integer to Roman
Pattern 89: String - String to Integer (atoi)	8. String to Integer (atoi), 65. Valid Number
Pattern 90: String - Multiply Strings (Manual Simulation)	43. Multiply Strings, 415. Add Strings, 67. Add Binary
Pattern 91: String Matching - Naive / KMP / Rabin-Karp	28. Find the Index of the First Occurrence in a String, 214. Shortest Palindrome, 686. Repeated String Match, 796. Rotate String, 3008. Find Beautiful Indices in the Given Array II
Pattern 92: String - Repeated Substring Pattern Detection	459. Repeated Substring Pattern, 28. Find the Index of the First Occurrence in a String, 686.Repeated String Match
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